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Design of a Variable Stiffness Unit for Prosthetic Sockets Based on Biomechanical Properties

Yawen Hu, Zexuan Zhou, Chunying Zou, and Li Jiang*

Abstract— **The socket serves as the physical interface connecting the prosthesis to the residual limb, with its performance directly influencing both prosthetic functionality and wearer comfort. Conventional sockets often employ rigid structures, leading to stress concentration causing soft tissue injuries, or creating voids between the socket and residual limb that compromise suspension performance. To enhance socket functionality and comfort, the study proposes a biomechanically compatible variable stiffness socket unit. The unit enables continuous adjustment of unit stiffness from 0.45 N/mm to 3.6 N/mm in response to external forces acting on the prosthesis, thereby dynamically optimizing interfacial pressure distribution. Based on the stiffness variation characteristics of human forearm soft tissues and their pain threshold distribution, the stiffness output range of the modular variable stiffness units was determined. The analysis provides a physiological rationale for the combinatorial design of these units. Furthermore, experimental evaluations were conducted to assess the output stiffness range and suspension performance of the modular units.**

I. Introduction

The prosthetic socket functions as the critical physical interface connecting the prosthesis to the residual limb. Its core roles encompass the secure and reliable transmission of force and motion [1], [2] along with providing a bidirectional neural interface for advanced prosthetic control [3]. Consequently, socket performance directly governs prosthetic functionality and wearer comfort.

Conventional sockets encapsulate the residual limb using rigid materials (sometimes with flexible inner liners) [4]. The static shape and mechanical properties of these rigid structures cannot adapt to the dynamic biomechanical behavior of soft tissues, frequently precipitating two clinical sequelae: First, during functional loading (e.g., grasping heavy objects), stiffness mismatch at the human-machine interface induces stress concentration. This results in localized contact pressures exceeding tissue tolerance thresholds, causing ischemic injuries. Second, persistent mechanical loading from high-stiffness designs during non-task periods accelerates muscle atrophy and induces neuropathic compression pain. Simultaneously, poor thermal dissipation [5] within the enclosed microenvironment promotes dermatological issues including edema and bacterial infections [6]. Frame-type sockets [7], [8] partially mitigate perspiration and heat accumulation issues. However, their characteristic pressure maldistribution frequently causes pain, pressure ulcers, and even soft tissue atrophy as adaptive responses [9]–[11].

To resolve this fundamental conflict, the study employed biomechanical testing to quantify forearm soft tissue stiffness variability and pain threshold distribution patterns. This underpins the design of a continuous-stiffness-adjustable socket unit capable of modulating its stiffness response during different operational states. This adaptation optimizes force and motion transmission under functional demands while satisfying physiological constraints. Furthermore, pain threshold distribution data is leveraged to develop a combinatorial unit strategy enabling dynamic switching between different unit combinations dynamically. This actively alleviates localized soft tissue pressures, mitigates fatigue or discomfort, and enhances overall safety and comfort. This work transcends the limitations of conventional static socket mechanics, establishing a novel paradigm for dynamic biomechanical adaptation within the prosthesis-human interaction system.

The paper is structured as follows: Section II summarizes the evolutionary trajectory of upper limb prosthetic sockets and introduces representative socket designs. Section III details the biomechanical characterization of human forearm soft tissues. Section IV presents the combinatorial design methodology for modular variable stiffness units. Section V describes the unit structural design and experimental analysis. Section VI concludes with principal findings and suggests future research directions.

II. Analysis of Current Research Status

Early socket designs only had a single suspension function, such as the full-contact Muenster socket shown in Figure 1, which achieved stable contact by fully contacting and closely fitting the residual limb. Poor heat dissipation, limited mobility, and stress concentration at the distal end of the residual limb led to the exploration of frame-type designs. The 3/4-type socket shown in the Figure 2 was proposed, featuring an open structure to enhance breathability and joint mobility [12]. Compared to lower limb prosthetic sockets, which prioritize load-bearing performance and gait stability, upper limb prosthetic sockets must meet the dexterity requirements for manipulating a prosthetic hand, emphasizing accuracy and flexibility in force and motion transmission, with relatively lower demands on load-bearing performance.

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The corresponding author is Li Jiang (jiangli01@hit.edu.cn).

Yawen Hu, Zexuan Zhou, Chunying Zou, and Li Jiang are with the State Key Laboratory of Robotics and System, Harbin Institute of Technology, Harbin 150001, China.

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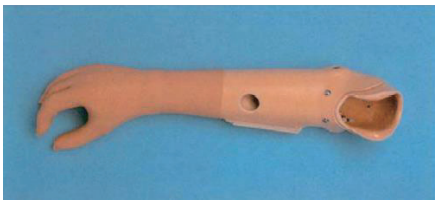


Figure 1. Muenster socket.

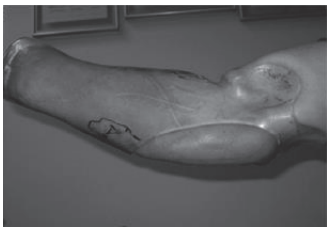


Figure 2. The 3/4-type socket.

Subsequent research shifted toward the integration of anatomical adaptation and dynamic function. As shown in the Figure 3, the variable suspension system uses silicone sleeves to switch between daily and sports scenarios, but material technology limitations prevented the resolution of thermal comfort issues [13]. As shown in Figure 4, the ACCI receiving cavity innovatively extends the fixation point to the proximal anatomical structure, optimizing axial rotation control and load distribution during movement, marking a refinement in biomechanical adaptation. The TRAC interface further integrates flexible materials with anatomical contours, enhancing elbow joint mobility while retaining the advantages of full contact, and incorporates optimization techniques from the 3/4-type cavity's olecranon region.

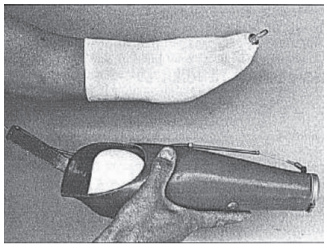


Figure 3. High-performance variable suspension receiving cavity.



Figure 4. ACCI socket.

Recent developments have focused on dynamic interaction mechanisms. High-fidelity interfaces (as shown in Figure 5) are based on Compression-Release Stabilization Theory (CRS), which uses zone pressure control to achieve residual limb bone movement control and load distribution, but there is a lack of systematic methods for individualized design. Researchers built upon this by integrating airbags and dynamic straps to enhance suspension adaptability and pronation functionality, while the negative stiffness pneumatic frame design regulates the compression state to balance safety, comfort, and force transmission efficiency, as shown in Figure 6 [14]. The core contradiction in current research has shifted to how to synergistically address the multidimensional demands of mechanical stability, anatomical adaptation, dynamic functionality, and long-term wearing comfort.



Figure 5. High-fidelity interface.

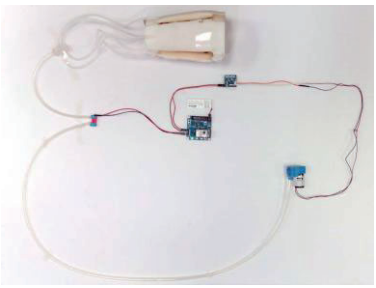


Figure 6. Negative stiffness pneumatic receiving chamber.

While existing socket designs have made significant progress in anatomical adaptation and dynamic functional control, they still face core challenges: the lack of quantitative data on soft tissue biomechanical properties limits the precision of socket mechanical adaptation. Especially in innovative designs such as high-fidelity interfaces and pneumatic dynamic suspension systems, key parameters such as contact force values and stiffness distribution still rely on empirical adjustments. This data gap results in a lack of theoretical support for variable stiffness mechanisms and individualized design relying on trial and error.

Precise quantification of residual limb biomechanical properties is a prerequisite for optimizing pressure distribution, yet the rigid structure of existing sockets struggles to adapt to the dynamic deformation demands of soft tissues during movement. Therefore, it is essential to establish stress-strain models of residual limb tissues under static and dynamic loads, quantify the mechanical response characteristics of soft tissues, and provide theoretical basis for acceptor cavity partition design. Additionally, developing reconfigurable stiffness adjustment modules that dynamically redistribute loads by adjusting unit stiffness can help resolve the inherent contradiction between stability and comfort in traditional designs.

III. BIOMECHANICAL EXPERIMENT ON HUMAN FOREARM

A. Subjects

The ideal subjects for forearm biomechanical experiments should be forearm amputees with varying degrees of residual limb conditions. However, the proportion of forearm amputees is relatively small, making it difficult to form a representative and comprehensive group of subjects. Therefore, the study enrolled a cohort of 10 healthy subjects (gender: 5 female/5 male; age: 23.9 ± 1.0 years) with no reported skin or musculoskeletal issues to investigate the distribution characteristics and biomechanical properties of the pressure pain threshold in the soft tissues of the forearm. Prior to the experiment, all participants were thoroughly informed of the study's purpose, procedures, and potential

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risks. For practical application, individualized adaptation should be performed based on the general patterns observed, while taking into account the specific anatomical and physiological characteristics of the residual limb. The study received ethical approval from the Ethics Committees of Harbin Institute of Technology, securing consent from all participants beforehand (ethical approval number HIT-2022023).

B. Protocol

A 4 mm diameter indenter with rounded edges was applied perpendicular to the skin surface at a speed of 1 mm/s. The edges of the indenter were rounded to avoid pain caused by stress concentration [15]. Experimental setup is shown in Figure 7, consisting of a hand-cranked loading test stand, a six-axis force sensor (Nano17), and a HEIDENHAIN length gauge. The displacement probe is connected to the force sensor to measure the displacement in the normal direction of the forearm's ventral plane. Combined with the normal force data collected by the sensor, this enables the calculation of stress and strain perpendicular to the forearm's plane.

The test position (using the right forearm as an example) is shown in Figure 8. The origin is set at the midpoint of the elbow crease, with the positive direction of the vertical axis extending along a straight line toward the midpoint of the wrist crease, and the positive direction of the horizontal axis extending from the origin toward the radial side. Test points are established at intervals of 45° in a counterclockwise direction (from the wrist toward the elbow), totaling 8 points, to form the coordinate system shown in the figure. Each point is measured three times, and the average of the three results is used as the final measurement value for that point.

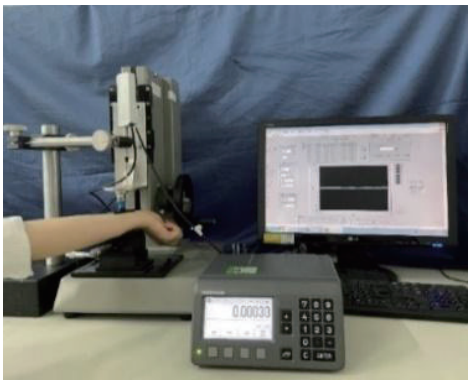


Figure 7. Experimental platform.

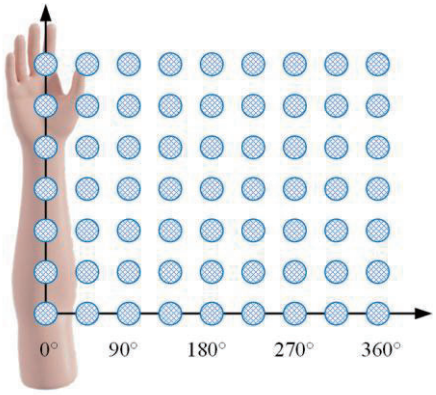


Figure 8. Forearm measurement coordinate system.

C. Data Analysis

Two indicators were selected: the pressure pain threshold (PPT), defined as the interfacial contact pressure (force per unit area) at which the subject initially perceives pain; and the deformation pain threshold (DPT), corresponding to the indenter displacement at pain onset [16]. A spatially resolved pain threshold matrix was generated by averaging all participants' measurements across the forearm coordinate system, yielding a population-averaged pain threshold distribution matrix [17].

Analysis of the human biomechanical experimental data presented in Figure 9 to Figure 11 yielded the following design guidelines for variable stiffness unit distribution:

- Gender exhibited no statistically significant impact on pain thresholds. However, females demonstrated lower mean pressure pain thresholds than males. This indicates potentially greater design complexity for female prosthetic users.
- Left and right forearms showed comparable deformation tolerance, supporting consistent unit distribution patterns across bilateral sockets.
- Proximal forearm regions tolerated higher pressure and deformation loads than distal regions. Consequently, units positioned proximally may safely apply marginally greater interface forces than distal units.

Simultaneously, the soft tissue stiffness change curve shown in Figure 12 reveal two distinct mechanical regimes: During initial loading, tissues display low stiffness (shallow curve slope), enabling large deformations under small stresses. With increasing load, tissue stiffness rises dramatically (steepening slope), requiring significantly higher stress to achieve further deformation.

Based on the tissue stiffness characteristics [18], [19], the design stiffness range for socket-tissue contact area was set as 5–15 N/mm. This range prevents pain from excessive pressure while avoiding insufficient pressure that compromises prosthetic suspension. Accordingly, the corresponding pressure required to induce tissue deformation was determined to be 2.5 N to 12 N. With the initial overlap distance between the socket unit and the skin set to 8 mm when not in contact,

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the unit stiffness range was calculated to be 0.48 N/mm to 3.64 N/mm. For practical purposes, the output stiffness of the unit was taken as 0.45 N/mm to 3.6 N/mm.

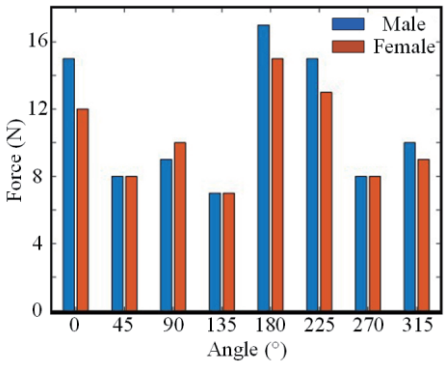


Figure 9. Comparison of pain thresholds between different genders.

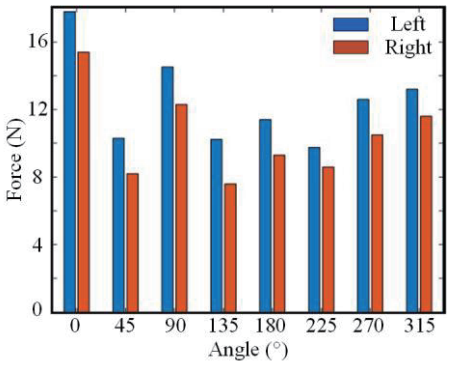


Figure 10. Comparison of pain thresholds between different sides.

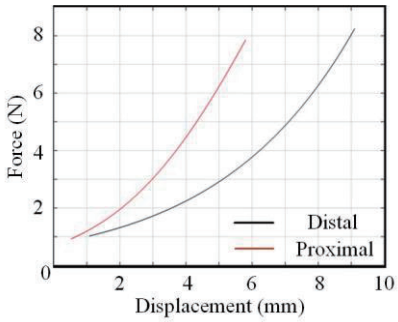


Figure 11. Proximal and distal stress-strain curves.

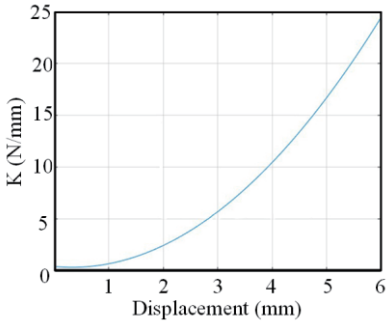


Figure 12. Soft tissue stiffness characteristics.

IV. Unit Combination Scheme Design

The combinatorial strategy for variable stiffness socket units adheres to four biomechanical principles:

- Avoidance of sensitive regions (i.e., areas with low pain thresholds) based on Section II experimental data;
- Unit quantity ensuring static equilibrium conditions under arbitrary external loads.
- The ability to dynamically regulate loads (compression or relaxation) on different regions of the forearm by switching between different unit combination states.
- While ensuring the accuracy of force and motion transmission of the receiving cavity, reduce the contact pressure between soft tissue and the unit surface.

Based on the above principles, the study adopts the strategy of arranging variable stiffness units along two cross-sections of the forearm axis, as shown in Figure 13, with the cross-sectional spacing depending on the length of the residual limb. The dual-cross-section distributed layout significantly increases the number of contact points, and combined with a 45° phase offset design, the spatial density of contact points at the receiving cavity-tissue interface is doubled (from 4 points in a single cross-section to 8 points in a dual cross-section). This feature enables greater spatial adjustability in pressure distribution when switching between unit combinations, allowing active avoidance of high-sensitivity areas with low pain thresholds (such as nerve bundle pathways or bony prominences), thereby significantly reducing the risk of pain induced by localized high pressure. The eight independently controllable units across the dual cross-sections can generate a rich array of combination patterns. During operation, support stiffness can be selectively enhanced in target muscle groups while reducing stiffness in antagonist regions, enabling independent and precise regulation of the forearm flexor/extensor muscle groups and pronator/supinator muscle groups, aligning with biomechanical functional zoning requirements.

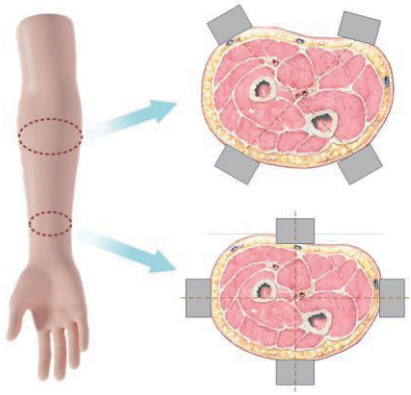


Figure 13. Schematic diagram of variable stiffness unit distribution.

The four units in the distal section are arranged in a circumferential orthogonal symmetrical pattern (0°, 90°, 180°, 270°), ensuring static load balance within a single section. The proximal section is rotated 45° relative to the distal section, forming an interlaced support structure in three-dimensional space. When any section unit experiences stiffness changes or failure, the other section can still provide complementary

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constraint forces, enabling the system to maintain excellent anti-overturning capability and overall stability under extreme or asymmetric external forces. The spatial interlaced design between units enables periodic alternation of tissue compression zones. During prolonged tasks, dynamic switching of activated unit combinations allows continuously compressed soft tissues to intermittently relax, preventing ischemic damage in a single region.

During non-task states, unit stiffness reverts to preset minima (0.45 N/mm) to maximize comfort. During task execution (such as manipulation or grasping), the stiffness of the units is continuously adjusted in response to changes in external forces to provide the necessary mechanical support. For tasks of longer duration, dynamic switching between different combinations of variable stiffness units can be achieved to alleviate local soft tissue pressure and reduce fatigue or discomfort.

V. VARIABLE STIFFNESS UNIT

A. Structural Design

Considering that the contact force required between the socket and the residual limb when amputees wear prostheses on a daily basis is not fixed but dynamically changes with limb movements and load variations, and significantly decreases when the arm is relaxed or at rest, the paper aims to design an adjustable stiffness socket unit that dynamically optimizes the contact force applied to the residual limb based on the external forces acting on the prosthetic hand. The design ensures the functionality of the socket while enhancing its comfort and safety.

The socket unit is based on the lever variable stiffness principle [20], utilizing a motor-driven screw mechanism to dynamically adjust the pivot point position, thereby altering the lever arm ratio between the input (spring) end and the output end, achieving continuous adjustment of output stiffness. The structure is shown in Figure 14. The lever structure adopts a plate design, with one end hinged to the upper housing of the unit and the other end connected to springs on both sides. The pivot point is fixed to the screw nut. When the motor rotates to drive the screw, it pulls the pivot point to move and simultaneously changes the spring deformation, thereby regulating the lever output load and ultimately achieving continuous adjustment of the system's overall output stiffness [21]. The variable stiffness unit is covered with a 1 mm thick silicone film [22], which not only enhances wearing comfort but also increases friction to achieve higher suspension performance. The dimensions of the variable stiffness unit are: 60 mm (length) × 30 mm (width).

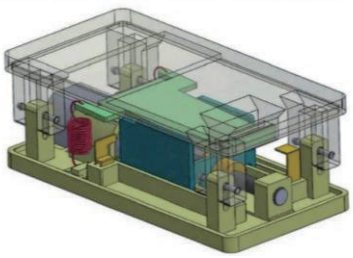


Figure 14. Variable stiffness unit sub-structure

B. Experimental Results and Analysis

According to the unit parameter design, the relationship between the theoretical output stiffness of the unit and the support displacement is shown below.

$$k_2 = k_1 \cdot \left(\frac{l + x}{L - l - x} \right)^2 \tag{1}$$

In the equation, k_1 is the input (spring) stiffness of the lever, k_2 is the output stiffness of the lever, L is the total length of the lever, l is the distance between the input end of the lever and the zero-reference position of the pivot, and x defines the displacement of the pivot point (i.e., linear offset from the zero-reference position).

The experimental platform for measuring the actual output stiffness of the unit is shown in Figure 15. One side of the variable stiffness unit is in contact with the fixed plane of the test bench, while the other side is in perpendicular contact with the six-axis force sensor. The distance moved by the unit after being subjected to pressure is read by the displacement meter, and the corresponding output force of the unit is read by the force sensor. The output stiffness of the unit is then calculated. Within the pivot point travel range ($x \in [0, 7.5$ mm$]$), the unit's output stiffness was tested in increments of 0.5 mm to determine how it changes with pivot point displacement.

The theoretical stiffness characteristics of the unit and the actual measurement results are shown in Figure 16. As shown in the figure, the output stiffness characteristics of the unit exhibit a trend similar to that of the soft tissue stiffness characteristics of the forearm, indicating that the receiving cavity unit can simulate the natural stiffness regulation mechanism of soft tissue. This enables the active matching of the prosthesis with the biomechanical characteristics of the human body. The convergence of the stiffness curves of the two components facilitates mechanical compatibility in dynamic loading conditions, preventing prosthesis slippage due to insufficient unit stiffness while avoiding circulatory disorders caused by excessive compression. This fundamentally optimizes fitting comfort and safety. The physiologically inspired variable stiffness design enables the socket unit to actively adapt to the biomechanical behavior of human soft tissue, laying a crucial foundation for achieving the functional integration of “human-machine unity.”

The primary function of the prosthetic socket is to ensure a stable connection between the residual limb and the prosthesis. Suspension performance serves as a key indicator of this stable connection, with the load on the prosthesis at the residual limb-socket interface when it reaches the critical state of sliding friction serving as a quantitative metric. In a limb suspension posture, test the static suspension capacity of the receiving cavity with different stiffness configurations for additional mass. Apply increasing tension along the forearm axis at the end of the prosthetic hand using a tension meter until relative slippage occurs at the interface, and record the critical load value.

In an upper limb suspension posture, when the unit stiffness is at its minimum value, the socket can stably support a 500g prosthetic hand self-weight without relative displacement at the interface, with a critical slip-off load

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threshold of 800g; when the unit stiffness is at its maximum value, the critical slip-off load threshold increases to 1400g.



Figure 15. Unit output stiffness test bench.

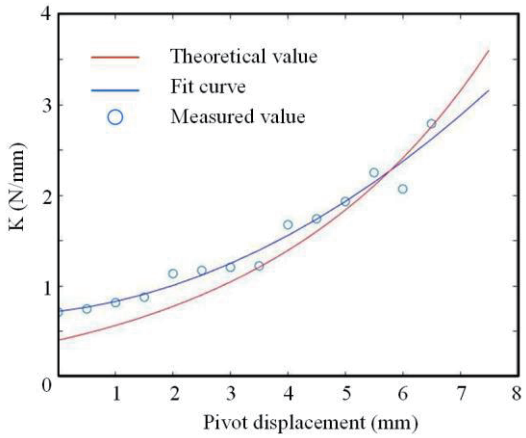


Figure 16. Unit theoretical stiffness and actual stiffness.

VI. Conclusion

The study proposes a biomechanically compatible variable stiffness receiving cavity unit that can continuously adjust the unit stiffness according to the external forces exerted on the prosthetic hand, thereby dynamically optimizing the pressure at the interface. Through human biomechanical experiments, the stiffness variation characteristics of forearm soft tissues were obtained, and the stiffness output range of the modular variable stiffness unit was determined. Based on the distribution characteristics of pressure pain thresholds and deformation pain thresholds, sensitive areas with lower pain thresholds were avoided, and the anatomical anterior, posterior, medial, and lateral regions were selected as the unit's active zones. This provides an important theoretical foundation for designing high-performance, high-comfort prosthetic receptacle units.

Future work will focus on achieving “dynamic response, zone optimization, and biomimetic adaptation” of the socket unit's force output and stiffness, thereby enhancing the accuracy of force and motion transmission and biomechanical compatibility.

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